

Analog Lab 4

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Objective :-

The aim of this experiment is to design two bandpass filters.

Where each

bandpass filter is designed to respond to a desired frequency.

We will derive

the values of resistors used and simulate the circuit on LTspice then will make a

physical circuit on breadboard to verify our output.

Required Material:-

1. MCP6004 as OPamps
2. Capacitors and Resistors
3. Power supply and Sinusoid wave (Adalm 1000)
4. Wire breadboard etc.

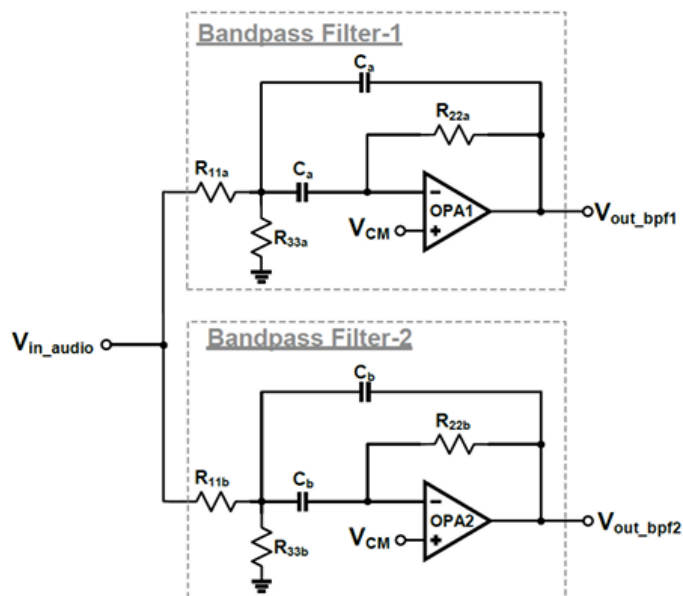
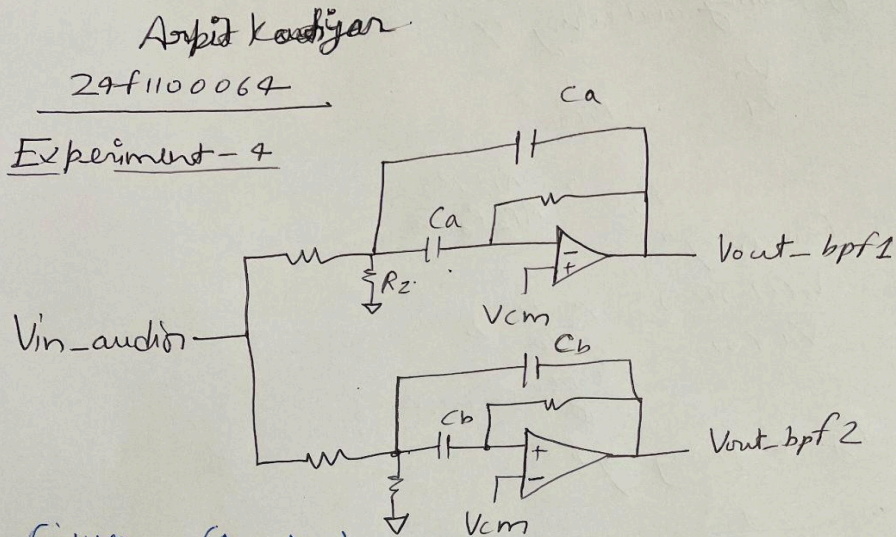


Figure 8 Schematic diagram of a bandpass filter.

Derivation:-



Given $(A_{01} - A_{02}) = 1$ and $Q_{01} = Q_{02} = 10$.

$[R_{11a} = R_1, R_{22a} = R_2, R_{33c} = R_3]$

$$\text{eq } 1 \quad \frac{V_i - V_x}{R_1} = \frac{V_x}{R_3} + \frac{V_2 - V_a}{1/C_1} + \frac{V_2 - V^-}{1/C_2}$$

$$\text{eq } ii \quad \frac{V_2 - V^-}{1/C_1} = \frac{V^- - V_0}{R_2} \quad [V^- = 2.5]$$

$$(V_x - V^-)C_3 = \frac{V^- - V_0}{R_2}$$

$$V_x C_3 - V^- C_1 = \frac{V^- - V_0}{R_2}$$

$$V_x C_1 = \frac{V^- - V_0}{R_2} + V^- C_3$$

$$V_x = \frac{1}{C_3} \left[\frac{V^- - V_0}{R_2} + V^- C_1 \right]$$

Put the V_x in eq i &.

By putting and solving eqⁿ

$$\frac{V_o}{V_i} = \frac{-\frac{1}{R_1 C} s}{s^2 + \frac{2}{R_3 C} s + \frac{R_1 + R_2}{R_1 R_2 R_3 C^2}}$$

Given eqⁿ is

$$\frac{V_o(s) - H(s)}{V_i(s)} = \frac{A_0 \omega_0}{s^2 + \omega_0 s + \omega_0^2}$$

By comparing eqⁿ

$$\omega_0 = \frac{1}{\sqrt{(R_1 R_2) R_3 C}}$$

$$R_1 = \frac{R_3}{2k}$$

$$R_2 = \frac{R_3}{4C^2 - 2k}$$

$$R_3 = \frac{R_1 C}{\omega_0 C}$$

Now let's find the value at R_1 , R_2 and R_3

For this I assume $C = 1 \mu F$

$$\omega_0 = 2\pi f = 2\pi \times 156.25 \\ = 981 \text{ rad/sec}$$

$$R_3 = \frac{2 \times 10}{981 \times 10^6} = 20 k\Omega \\ (\text{exact value} = 20.32 k\Omega)$$

$$k = 1 \quad R_1 = \frac{20 k}{2(1)} = 10 k\Omega \\ (\text{given}) \quad (\text{exact value} = 10.19 k\Omega)$$

$$R_0 = \frac{20}{4 \times 100 - 2} = \frac{50.25 \Omega}{} = 50 \Omega$$

Similarly we find for 2nd filter and here I assume $C = 0.1 \mu F$ by same cal. I got .

$$R_1 = 25 k\Omega$$

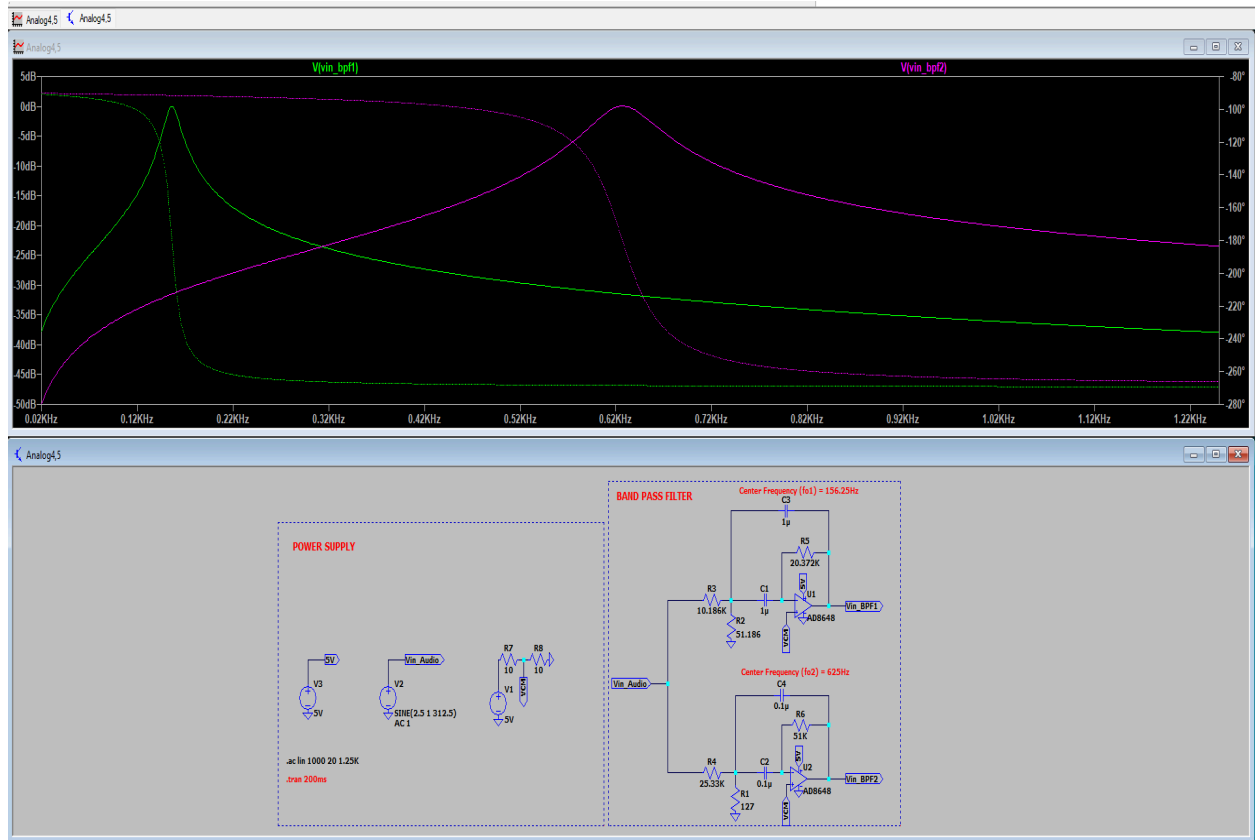
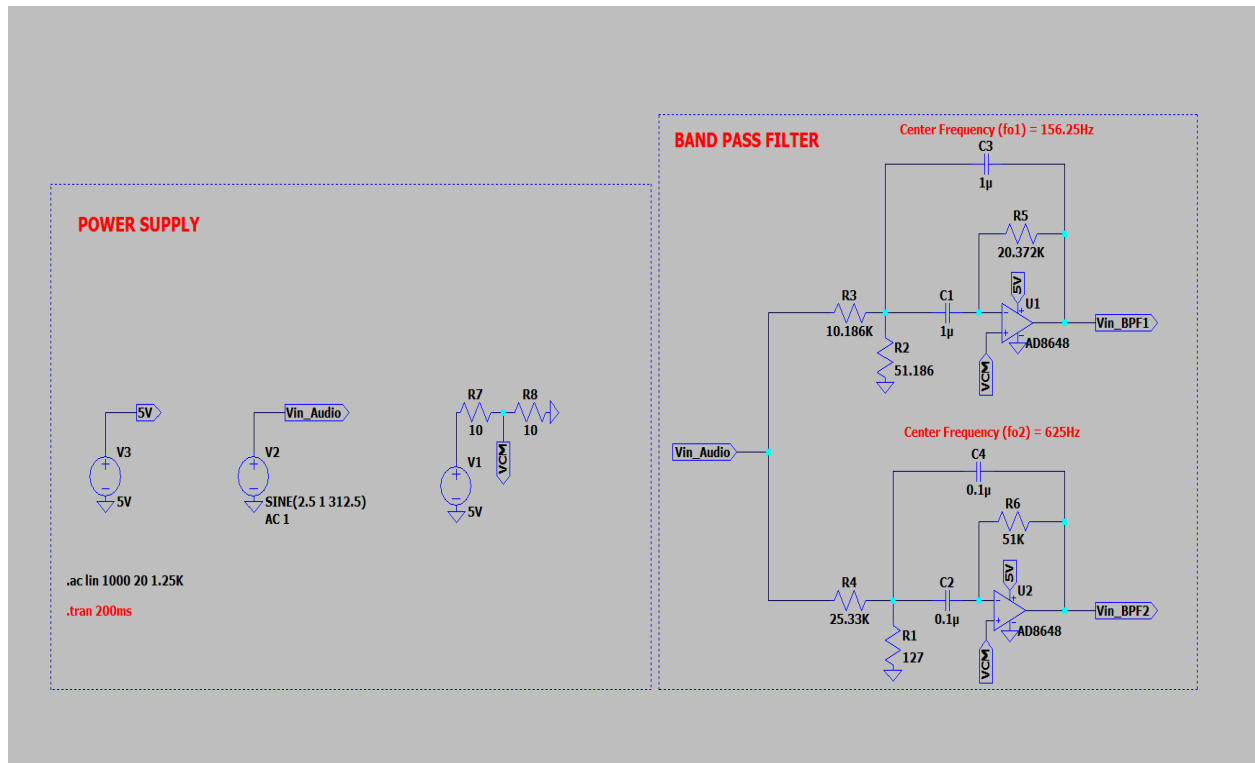
(exact value = 25.33 k Ω)

$$R_2 = 50 k\Omega$$

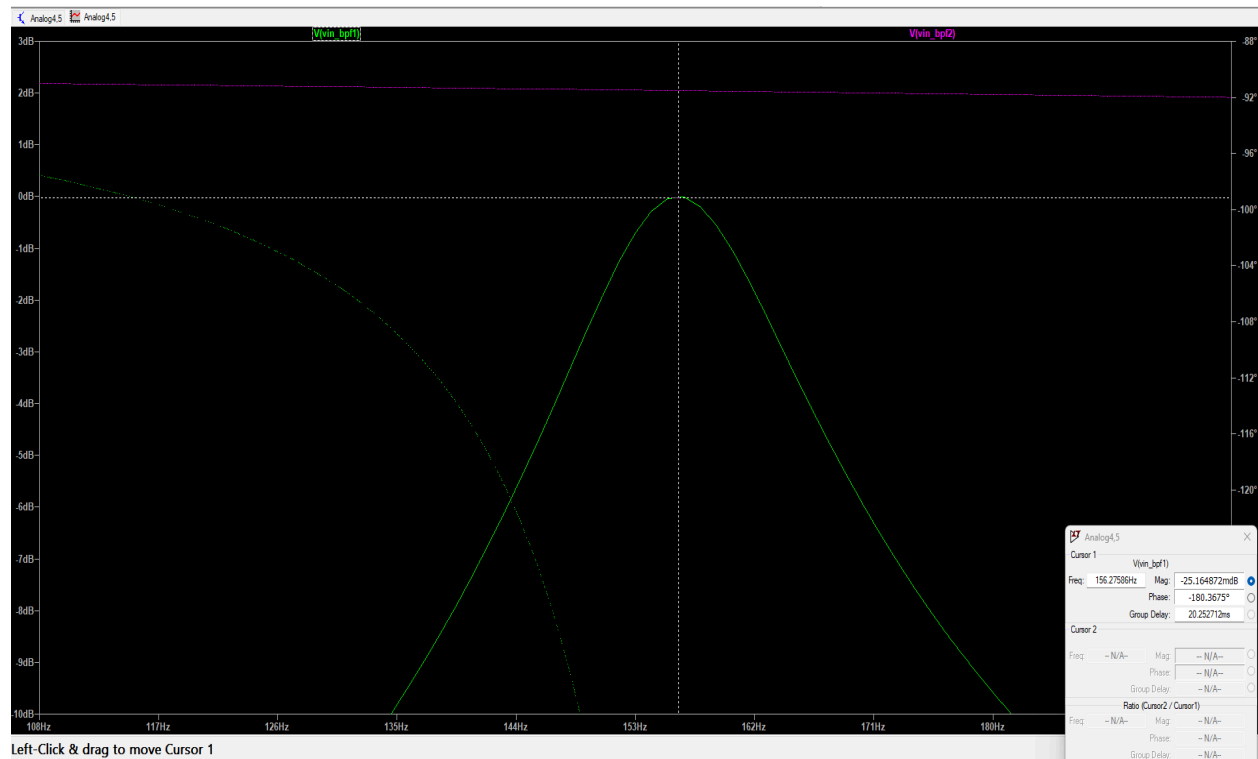
$$R_3 = 12 \Omega$$

This value I am only use in physical circuit

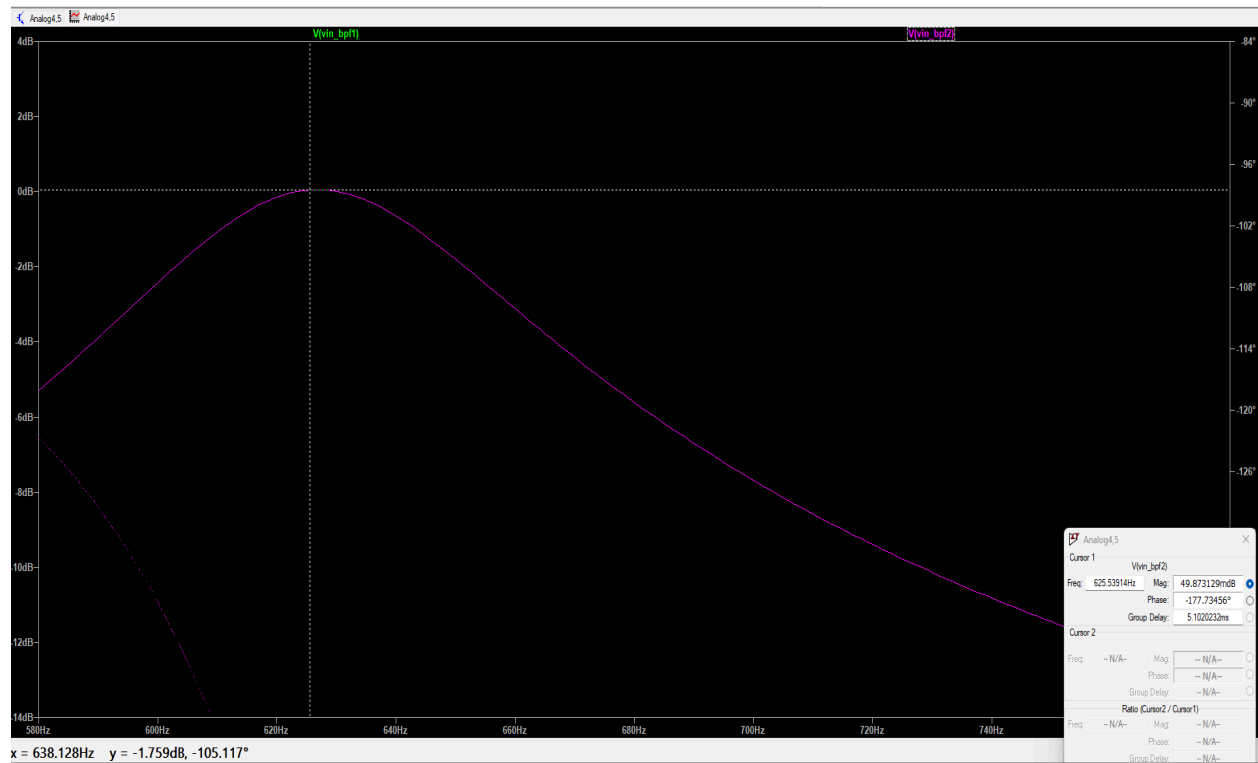
LTSpice Circuit:-



BandPass Curve With 156.25Hz Cutoff Freq



BandPass Curve With 625Hz Cutoff Freq



BandPass Filter Bode plot With 156.25Hz and 625Hz Cutoff Freq

